

Title: Constructing a Pokémon World: Analyzing Language, Sentiment, and Type Across Generations

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Goal Statement: We will analyze Pokémon descriptions across generations to identify how descriptive vocabulary and sentiment vary by Pokémon type and generation, thereby revealing patterns in the language used to build a cohesive fictional world.

Research Question: How can Pokémon's approach to worldbuilding inform other game developers seeking to build a cohesive franchise?

Narrative Paragraphs

Pokémon's 30-year-old universe, spanning over 1,028 species [1], feels like one cohesive world because of the consistent tone used in writing Pokémon species entries. We are analyzing the explicit descriptive vocabulary of Pokédex entries across all nine generations to determine how developers used language to construct a distinct fictional reality for each Pokémon type. This matters because that same cohesiveness of Pokémon's multi-generational franchise can help developers achieve something similar when building their own IPs. Furthermore, fans creating their own Pokémon species, either to share with the community or for their own fan games, could benefit from referencing this existing worldbuilding template.

Currently, Pokémon's mechanical data, like types, stats, and movesets, are well documented through APIs like PokeAPI, but the prose of Pokédex entries has not been studied to the same degree. Prior work [2] has used Pokédex text to predict a Pokémon's type with one-hot encoding, GloVe, and transformers, but that work treated the text only as input for a prediction task. Our project shifts the focus from prediction to understanding the text as a worldbuilding mechanism. By identifying patterns in vocabulary and sentiment across Pokémon types and how this differs across generations, we aim to show how language helps maintain story cohesion in a decade-spanning franchise.

Modeling Approach

We will use hierarchical clustering to identify patterns in the vocabulary and sentiment of Pokémon species entries across different Pokémon types and generations. Specifically, we will examine relationships between descriptive words and their associated sentiment scores to identify shared linguistic characteristics between Pokémon types, but also within-type variations across generations. Hierarchical clustering is an “unsupervised machine learning” [3] approach that detects similar objects and clusters them based on their degree of dissimilarity. This approach is appropriate because it allows exploring naturally occurring relationships in our data without assuming beforehand what groupings will emerge. It also allows us to investigate potential language hierarchies associated with different Pokémon types. For example, we may find that Fire and Electric types share similar descriptive patterns, while Ground and Rock types exhibit their own unique degrees of similarity. These clusters further enable us to identify how linguistic and emotional characteristics have developed across generations.

References

- [1] "Pokémon (species)," *Bulbapedia, the community-driven Pokémon encyclopedia*. [Online]. Available: [https://bulbapedia.bulbagarden.net/wiki/Pok%C3%A9mon_\(species\)](https://bulbapedia.bulbagarden.net/wiki/Pok%C3%A9mon_(species)). [Accessed: Sep. 9, 2026].
- [2] R. Belfer, "Predicting Pokémon Type Using Pokédex Entries," *Medium*, Jun. 11, 2021. [Online]. Available: <https://medium.com/analytics-vidhya/predicting-pok%C3%A9mon-type-with-the-pok%C3%A9dex-7038754dc422>. [Accessed: Sep. 9, 2026].
- [3] J. Noble, "What is hierarchical clustering?," *IBM*, [Online]. Available: <https://www.ibm.com/think/topics/hierarchical-clustering>. [Accessed: Sep. 9, 2026].